

XGBoost-based Solar Forecasting for the Galapagos Islands

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Abstract. Isolated island microgrids historically rely on unstable, carbon-intensive thermal energy generation via fossil fuel imports. Integrating renewable energy resources can mitigate these ecosystem risks, yet high short-term meteorological variability complicates grid stabilization. This paper presents a solar nowcasting framework trained on weather station data, utilizing a 49-feature architecture engineered entirely from pre-existing ground meteorological instruments. We deploy six independent Extreme Gradient Boosting (XGBoost) regression models across 30 to 180-minute horizons. Experimental results demonstrate that the framework significantly outperforms standard persistence and climatology benchmarks across all horizons, yielding a Mean Absolute Error ranging from 34.57 W/m² at +30 minutes to 61.74 W/m² at +180 minutes. Feature importance evaluations verify that the underlying machine learning models base their predictions on physically coherent dynamics. This approach provides a highly interpretable, hardware-efficient, and replicable forecasting strategy capable of facilitating renewable energy integration.

Keywords: Solar forecasting · Nowcasting · XGBoost · Machine Learning · Renewable Energy · Island microgrids.

1 Introduction

The Galapagos archipelago, declared a UNESCO World Natural Heritage Site in 1978, faces a fundamental tension between ecological preservation and growing energy demand. Due to its isolation from the Ecuadorian mainland, the islands are not connected to the national grid and rely entirely on isolated local generation. Historically, thermal power plants supplied the majority of this electricity,

necessitating heavy imports of fossil fuels, and posing severe risks to the local ecosystem through carbon emissions and potential oil spills, with significant maritime accidents having occurred [10].

Over the last two decades, the islands have transitioned toward renewable energy resources, primarily wind and solar power. Despite the growing deployment of renewable infrastructure, accurate short-term solar radiation forecasting remains an unresolved challenge for the island’s energy system. Without reliable nowcasting tools, operators lack the information needed to anticipate changes in solar generation, limiting their ability to optimize the balance between renewable and thermal sources [9, 7]. The existing solar forecasting literature offers limited guidance for this context. Most prediction models have been developed and validated for utility-scale, grid-connected installations in continental settings [2, 6]. Island microgrids operating in isolation, with unique meteorological conditions, remain largely unaddressed [3].

The main contributions of this paper are twofold: first, we introduce a solar nowcasting system trained strictly on meteorological data, demonstrating that accurate short-term forecasting is achievable using existing ground infrastructure. Second, we develop a six-horizon XGBoost model (30 to 180 minutes ahead) that substantially outperforms traditional baselines, yielding a mean absolute error ranging from 34.57 W/m^2 at the 30-minute horizon to 61.74 W/m^2 at the 180-minute horizon.

2 Related Work

Early approaches to solar forecasting relied on statistical and persistence-based methods. The persistence model assumes that irradiance at time $t+h$ equals the value observed at time t ; it remains the standard naive baseline in solar forecasting due to its simplicity [14]. Autoregressive models such as ARIMA extended this approach by capturing temporal autocorrelations, but their reliance on linear assumptions limits their capacity to model the nonlinear dynamics governing cloud formation and dissipation [12]. The emergence of machine learning (ML) brought substantial improvements. Support vector machines, random forests, and gradient boosting methods demonstrated superior accuracy over statistical baselines across diverse climatic conditions [1].

3 Data and Study Area

3.1 The Weather Station Network

San Cristóbal Island is monitored through a local weather station network distributed across different elevations, from lowland areas to the island highlands. As shown in Fig. 1, the network includes four meteorological stations: El Mirador, Merceditas, Cerro Alto, and El Junco. The Galápagos Science Center (GSC) is also shown in the map as a geographical reference point, but it is not considered part of the weather station network used in this study.

This study focuses exclusively on the El Junco station, located in the high-land region of San Cristóbal at approximately 700 m above sea level. El Junco was selected because, among the available station datasets, it showed the lowest proportion of missing values for the key variables required by the forecasting framework, including precipitation, temperature, relative humidity, solar radiation, wind speed, and wind direction.



Fig. 1. Spatial distribution of the weather station network on San Cristobal Island [11]

3.2 Dataset Description

The dataset consists of continuous observations spanning more than 10 years, from June 2015 to March 2026. At the original 15-minute temporal resolution, the series contains approximately 376,000 raw records, capturing both diurnal variance and long-term seasonal cycles.

To prevent data leakage, the time series was partitioned chronologically. The training set spans from June 2015 to January 2024 (150,087 timesteps), and the testing set encompasses January 2024 to March 2026 (37,522 timesteps) reserved exclusively for model validation. The feature vector incorporates 14 variables, including incident solar radiation, air temperature, relative humidity, wind speed, wind direction, and cumulative precipitation.

4 Methodology

4.1 Preprocessing and Signal Smoothing

Raw 15-minute meteorological measurements from El Junco were resampled to a 30-minute frequency by interval averaging. This scales the time series to

match our multi-horizon targets (30 to 180 minutes) while smoothing out micro-fluctuations. As illustrated in Fig. 2, the 30-minute resampling filters out sudden mid-day spikes caused by fleeting cloud cover while fully preserving overall diurnal trends.

Following feature extraction, any incomplete rows caused by rolling windows or historical lags were dropped rather than interpolated to maintain physical integrity. Lastly, data was restricted strictly to daylight operational hours to prevent nighttime zeros from biasing model training.

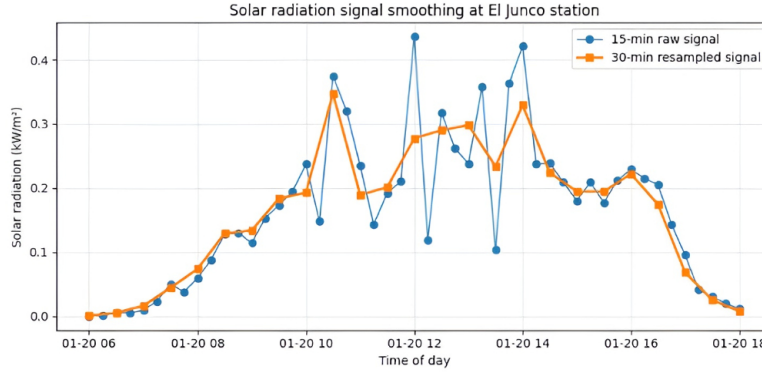


Fig. 2. Effect of 30-minute resampling on the solar radiation signal at El Junco station.

4.2 Feature Engineering

The feature engineering framework converts raw meteorological and operational variables into predictors by embedding physical knowledge directly into the dataset. These engineered features are presented in Table 1. To capture temporal patterns without introducing artificial step discontinuities, hours and days-of-year are mapped to continuous cyclical sine and cosine components. Localized atmospheric dynamics and volatility are quantified using the velocity, acceleration, and rolling variance of solar irradiance. Air mass transitions, condensation risk, and surface moisture persistence are tracked through humidity derivatives, a joint humidity-solar risk index, and a 6-hour delayed leaf wetness indicator. To maintain continuous spatial tracking of weather, wind speed and direction are decomposed into orthogonal vector components. Finally, daily persistence trends are captured through 6-hour and 24-hour lags, while short-term effects utilize immediate histories (30-to-180-minute steps) and true net hourly changes.

After feature engineering, the dataset consisted of 49 features.

4.3 The Meteorological Predictor

The framework utilizes the Extreme Gradient Boosting (XGBoost) algorithm, an ensemble of sequential, additive regression trees where each tree corrects

Table 1. Feature Engineering Categories and Definitions for Solar Irradiance Forecasting

Category	Feature Name(s)	Definition
Cyclical time	hour_sin, hour_cos, doy_sin, doy_cos	sin, cos of hour (24h) and day of year (365.25d)
Ramp mechanics	v_solar, a_solar, solar_volatility_2h	1st/2nd derivative of solar irradiance; 2h rolling SD
Environmental risk	v_humidity, humidity_solar_risk	1st derivative of humidity; joint humidity-solar index
Lag memory	{sensor}_lag_6h, {sensor}_lag_24h	6h ($t - 12$) and 24h ($t - 48$) delays for 7 sensors
Short-term history	solar_lag_30m to solar_lag_180m	History from $t - 1$ to $t - 6$
Clear sky proxy	solar_time_ratio	Solar irradiance / (hour_sin + 0.1), clipped to $[0, 5]$
Wind vector	wind_x, wind_y	Wind speed $\times$ cos/sin of wind direction radians
Wet ground memory	wet_ground_lag_6h	6h lag ($t - 12$) of leaf wetness sensor
Trend analysis	solar_trend_1h	Current solar irradiance minus 1h lag ($t - 2$)

the residual errors of its predecessors [4]. The model minimizes a regularized objective function combining a differentiable loss term with a complexity penalty to prevent overfitting.

Rather than predicting the absolute solar irradiance directly, each regressor is trained to predict the ramp: the change in solar radiation (ΔG) between the current time step and the target horizon. The final irradiance forecast is then reconstructed by adding the predicted ramp to the last observed value, focusing the learning process on highly variable cloud dynamics.

The framework employs six independent XGBoost regressors, one for each forecast horizon (30, 60, 90, 120, 150, and 180 minutes). Training a dedicated model per horizon allows the network to adapt to evolving predictor–target relationships as lead times lengthen. Models are optimized using a pseudo-Huber loss (`reg:pseudohubererror`) to reduce sensitivity to outliers caused by abrupt cloud changes. XGBoost was chosen for three key operational reasons: it achieves high predictive accuracy on tabular meteorological data without the heavy computational demands of deep learning [12, 1]; its light hardware footprint suits resource-constrained microgrids [13]; and it provides native feature importance tracking to ensure physical interpretability. This transparency is particularly valuable for building operator trust and for verifying that the model’s predictions are grounded in physically coherent relationships.

4.4 Baseline Models

To contextualize the performance of the proposed method, two standard reference models were evaluated under identical conditions. First, persistence is the canonical naive baseline in solar forecasting [14]. Persistence assumes that the irradiance at a future time $t + h$ equals the most recent observed value at time t , propagating the current value unchanged across all horizons. Second, climatology, also referred to as the historical mean, uses the historical average over the training set. By construction, it ignores the current atmospheric state and cannot react to day-specific conditions such as transient cloud cover.

5 Results and Discussion

5.1 Predictive Performance

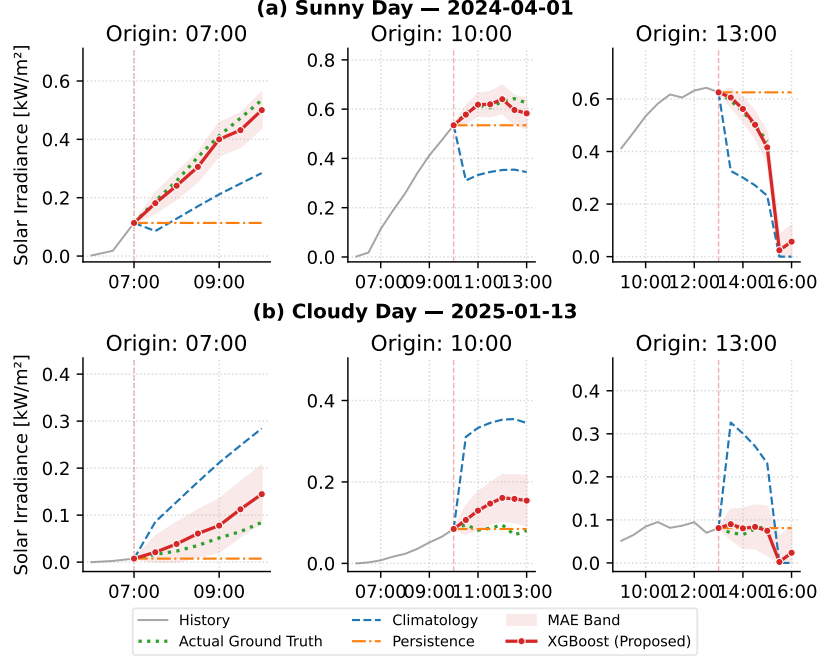
As shown in Table 2, the proposed XGBoost model outperforms both baselines across all horizons. While the XGBoost framework’s error increases by 79% moving from the 30-minute to the 180-minute forecast, the baseline persistence error degrades by 242%. The climatology baseline remains flat but exhibits unacceptably high static error margins. Similarly, the proposed method offers the lowest standard deviation.

Fig. 3 showcases the model and baselines performance for a sunny and cloudy day. In the first case, the model follows the ground truth closely throughout the day, achieving lower error than the other methods. On the contrary, the model

Table 2. Comparison of Mean Absolute Error and standard deviation (W/m^2)

Horizon	XGBoost	Persistence	Climatology
+30 min	34.57 ± 39.45	44.70 ± 44.35	80.85 ± 68.09
+60 min	45.77 ± 49.75	73.80 ± 63.55	81.43 ± 67.54
+90 min	50.21 ± 52.54	99.12 ± 76.54	82.82 ± 66.73
+120 min	56.56 ± 57.31	121.14 ± 87.78	85.22 ± 66.14
+150 min	58.23 ± 58.78	139.07 ± 98.66	88.99 ± 66.55
+180 min	61.74 ± 60.62	153.06 ± 107.40	94.02 ± 68.78

exhibits higher error in the low irradiance setting. This is attributed to the movement of clouds making the weather more unpredictable, and vastly different from the historical mean baseline. Despite this, the model is able to follow the general trend, and result in better performance than the persistence baseline.

**Fig. 3.** Solar irradiance forecasting performance under different atmospheric conditions: (a) Sunny day scenario, and (b) Overcast day scenario.

Residual error distributions are presented for each horizon model in Fig. 4. For every case, the error is centered closely around 0, with the error and standard deviation growing larger with a longer horizon. The model is able to accurately predict the solar availability for the short-term window, with a MAE of $35 W/m^2$.

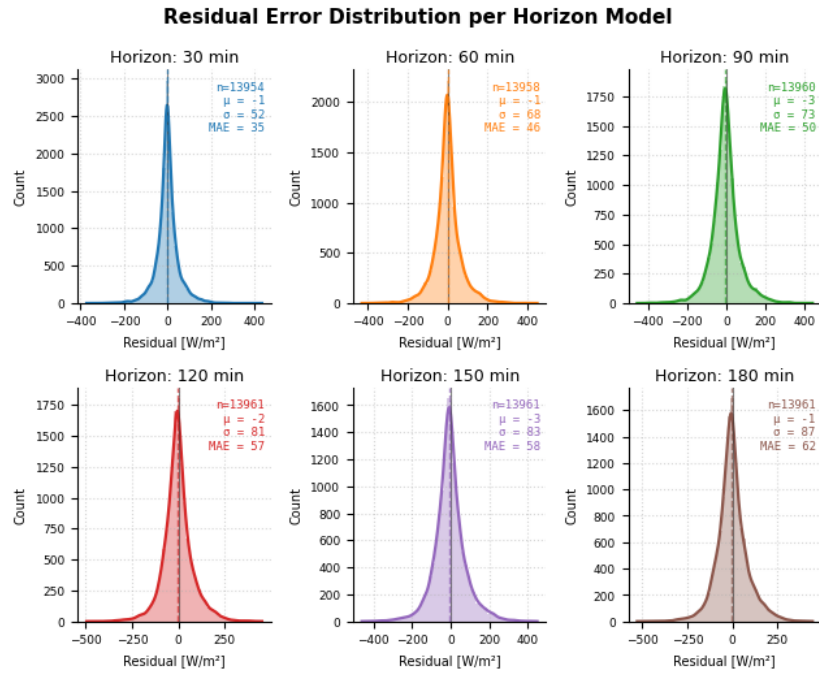


Fig. 4. Residual error distribution for the 6 XGBoost models. Each plot features the number of samples, mean (μ), standard deviation (σ), and MAE.

5.2 Model Explainability

Model explainability was analyzed through the feature importance estimated by the XGBoost models. The global importance of each feature was computed as the sum of its contributions across all models. Fig. 5 shows the fifteen most relevant features for predicting solar radiation changes at the El Junco station. The results show that cyclical temporal features, especially `hour_sin` and `hour_cos`, are the most important variables in the model. This indicates that the daily evolution of solar radiation is strongly conditioned by the time of day, as expected due to the solar cycle. Current and lagged radiation values, such as `solar_kw`, `solar_kw_lag_6h`, `solar_lag_180m`, and `solar_lag_150m`, also appear among the most relevant features. These variables reflect the temporal persistence of the solar signal and help capture the recent evolution of cloud cover and irradiance availability. `humidity_solar_risk` and `v_humidity` are also highly relevant, incorporating the effect of relative humidity on solar variability. Increases in humidity may be associated with cloud formation that reduce incoming radiation. Wind-related variables showed lower relative importance.

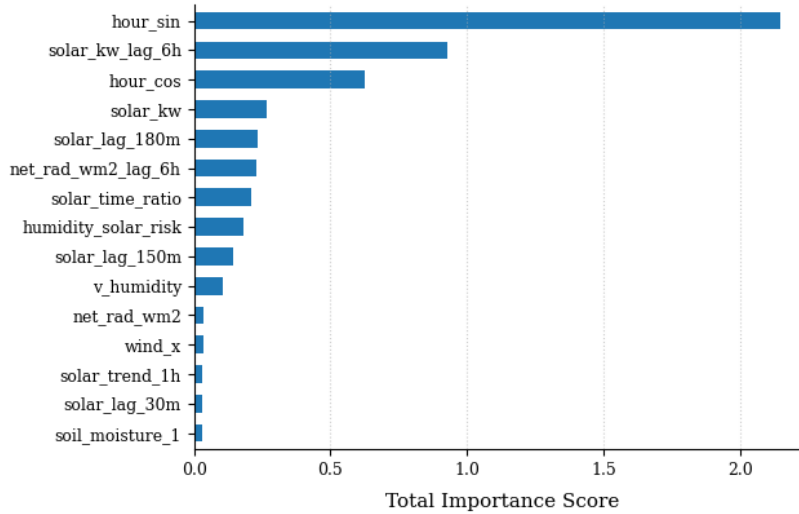


Fig. 5. Feature importance accumulated across the six XGBoost horizon models.

5.3 Applicability to Other Microgrids

The proposed methodology is not limited exclusively to San Cristobal Island or the Galapagos archipelago. The forecasting framework relies solely on meteorological variables commonly recorded by conventional weather stations; its implementation does not require costly, specialized infrastructure, facilitating its adoption for resource-constrained environments. The proposed approach is particularly relevant for isolated microgrids that combine renewable generation

with thermal backup generation. Such systems are common in small islands and remote communities seeking to reduce their dependence on fossil fuels through greater integration of renewable energy sources [5, 8]. Consequently, the methodology could be extended to other power systems.

6 Conclusion

This paper presented a solar nowcasting framework designed to support renewable energy integration in island microgrids. Using meteorological data from the El Junco summit station (700 m a.s.l.) on San Cristobal Island, Galapagos, we demonstrated that a set of six independent XGBoost regression models can achieve mean absolute errors ranging from 34.57 W/m^2 at 30 minutes to 61.74 W/m^2 at 180 minutes. These results substantially outperform the temporal mean baseline. The results demonstrate that accurate solar nowcasting can be achieved for isolated island environments using only a single existing ground weather station, without requiring satellite imagery, numerical weather prediction models, or additional sensor deployment. This finding is particularly relevant for small island developing states pursuing renewable energy transitions under resource constraints. It is important to emphasize that the scalability of the proposed approach lies in the methodology rather than in the specific models trained. Each new deployment would require a training process using local meteorological observations in order to capture the specific climatic and geographic characteristics of the target location. This study constitutes a replicable strategy that can facilitate renewable energy integration and accelerate the energy transition of isolated microgrids worldwide.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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